

INVITATION DETAILED: DEEP BIO ACCELERATOR FOUNDER INTERVIEW

Invitation received from the Accelerator Success Team at HomeLab (Powered By LabFellows) for the Founder Interview. Focus indicates Launch Track parameter alignment (\$0M - \$2M funding targets).

Host: HomeLab Team (startup@homelab.com) | Format: 30-Min Video Call (All Co-founders Required)

1. Executive Summary & Value Proposition

Nilus Lab is pioneering therapeutics in longevity by modeling generative cellular reprogramming trajectories to reverse cardiac ageing without tumor risk (oncogene-free). Our proprietary platform, Zenith v28.0, leverages a deep foundation manifold trained on 500,000+ single cardiac cells from the Human Cell Atlas (HCA) and the PERIHEART dataset. By utilizing variational autoencoders (scVI) and molecular docking simulations, Zenith identifies combinatorial gene networks that restore youthful chromatin profiles and restore contractility in senescent cells.

2. Technology & Platform Validation (Traction)

Generative scVI Core: Trained on 500k Verified Human Cardiac Cells, capturing 4,908 highly variable genes across 400 epochs.

Epigenetic Reset: Achieved a deterministic -11.9 year epigenetic age reversal in mature cells verified via molecular clocks.

Safety Gating: Built-in safety ceilings automatically filter out oncogenes and restrict dedifferentiation to protect functional cell identity.

AF3 Valency Audit: Integrates automated AlphaFold 3 structured motifs to verify target binding affinity (e.g., POU5F1-SOX2 complex).

3. Competitive Landscape

Retro Bio & Altos Labs: Focus on broad, systemic rejuvenation which requires hundreds of millions in funding. Nilus targets tissue-specific (cardiac) indications with lean, computational efficiency.

Turn Biotechnologies: Utilizes mRNA-based epigenetic restoration but focuses on dermatology and skin indications. Nilus is optimized specifically for the cardiac lineage.

The Nilus Edge: In-silico dosage networks predict transient transcription factor windows (e.g., 2h ON / 21h OFF) to resolve oncogene and cell drift risks before lab synthesis.

4. Founder Interview Preparation Guide

The Founder Interview is a highly focused 30-minute video call. The HomeLab team will look to evaluate the following core criteria. Anticipated talking points are structured below:

Technology Validation: Explain the transition from scVI model predictions to in-vitro validation. Highlight the use of real HCA datasets, chromatin-binding valency models, and Horvath Clock verification.

Market indications: Position Nilus against the \$50B longevity market. Initial clinical focus is on cardiomyopathy and age-induced heart failure, productized as an out-licensing platform for pharma.

Competition & Barriers: Demonstrate how Nilus bypasses Altman-style reprogramming risks (oncogenesis, cell drift) computationally, creating strong IP and high safety barriers compared to Altos/Retro.

Team Chemistry & Dynamics: Since all co-founders are required to join, show the interdisciplinary synergy across computation (VAE architectures), systems engineering, and molecular biology.

Accelerator Goals: Secure next round of funding (\$0M - \$2M Launch Track parameters), leverage HomeLab lab networks for early animal trials, and expand hospital partnerships for cell samples.

5. HomeLab Contact & Follow-up Details

The video call is scheduled using the booking link in the invite. In case of schedule shifts or coordinator follow-ups, the Accelerator Success Team coordinates via HomeLab Powered By LabFellows:

- Email contact: startup@homelab.com
- Tel contact: +1 (833) 452-2677 ext. 500
- HomeLab homebase website links local tools and preparation documents.